

Mobile colistin resistance (*mcr*) genes and recent developments in colistin resistance detection

Divya Lakshmanan^{1,*}, Dhamodharan Ramasamy¹, Veni Subramanyam¹, Suresh Kumar Saravanan²

¹Mahatma Gandhi Medical Advanced Research Institute (MGMRI), Sri Balaji Vidyapeeth (Deemed-to-be-University), Pillayarkuppam, Pondicherry 607042, India

²Mahatma Gandhi Medical Preclinical Research Centre (MGMPRC), Sri Balaji Vidyapeeth (Deemed-to-be-University), Pillayarkuppam, Pondicherry 607402, India

*Corresponding author. Mahatma Gandhi Medical Advanced Research Institute (MGMRI), Sri Balaji Vidyapeeth (Deemed-to-be-University), Pillayarkuppam, Pondicherry 607402, India. E-mail: divyal@mgmari.sbv.ac.in

Abstract

The peptide antibiotic colistin has been reserved as a last resort antibiotic treatment option for cases where other antibiotics including carbapenems have failed. Recent emergence of colistin resistance and discovery of mobile colistin resistance (*mcr*) genes, which encode the cell wall modifying phosphoethanolamine transferase enzyme, complicates the issue. The *mcr* genes have been associated with conjugative plasmids and can be horizontally transferred between different bacterial species. The global spread of *mcr* genes has been extensively documented and this warrants surveillance of the resistance genes in the community. However, susceptibility testing of colistin is fraught with practical challenges owing to the chemical nature of the drug and multiple mechanisms of resistance. Although broth microdilution is the current gold standard for colistin susceptibility testing, the method poses technical challenges. Hence, alternative detection methods for screening colistin resistance are the need of the hour. Several methods have been studied in the recent times to address this issue. In this review, we discuss some of the recent developments in the detection of colistin resistance.

Keywords: colistin resistance, mobile colistin resistance (*mcr*) genes, antimicrobial susceptibility testing, resistance detection

Introduction

Multidrug resistance has become a matter of grave concern in the recent times. The emergence of extensive drug resistance and pan-drug resistance has left the medical fraternity searching for alternative treatment options. According to World Health Organization (WHO) (2023), nearly 5 million deaths in 2019 were associated with antimicrobial resistance (AMR). Emergence of carbapenem resistance among Gram-negative bacteria has pushed clinicians to use antibiotics such as colistin as a last resort. Polymyxins, especially colistin, are associated with nephrotoxicity and has limited bioavailability in lungs, making them salvage treatment option (Satlin et al. 2020). However, colistin resistance is also being increasingly reported from different countries (<https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>). Multiple genes confer resistance to colistin; the most recently identified among them are the mobile colistin resistance (*mcr*) genes. There are 10 variants of *mcr* genes known till now (Hussein et al. 2021), which are encoded in plasmids and can be horizontally transferred.

Colistin susceptibility is assessed through broth microdilution (BMD) as per Clinical and Laboratory Standards Institute (CLSI) and European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines. However, interpretation of colistin minimal inhibitory concentration (MIC) poses practical challenges due to its chemical properties. BMD is cumbersome for routine testing and determining MIC is tough especially with bacteria like *Acinetobacter* and *Pseu-*

domonas. However, monitoring colistin resistance is crucial due to rising cases of resistance. Thus, there's a pressing need for reliable and swift detection methods. In this review, we highlight the recent developments in detection of colistin resistance.

Colistin—mechanism of action and resistance

The outer Lipopolysaccharide (LPS) layer of Gram-negative bacteria protects them from harmful compounds, including antibiotics. A specific lipid moiety, 'Lipid A', acts as a surface anchor to the LPS of Gram-negative bacteria. Colistin (Polymyxin E) is a polypeptide cationic antibiotic, which binds to the anionic lipid A of the LPS layer by electrostatic interaction and displaces the Ca^{2+} and Mg^{2+} ions. This leads to the disintegration of the outer membrane, its permeabilization and subsequent leakage of cell components, and death (Gogry et al. 2021). Resistance to colistin occurs by different mechanisms (Fig. 1), the most common one being the modification of lipid A component by addition of phosphoethanolamine (PEtN) or a 4-amino-4-deoxy-L-arabinose (L-Ara4N) molecule. This renders the outer membrane less anionic and prevents binding of colistin (Gogry et al. 2021). The addition of phosphoethanolamine is carried out by the phosphoethanolamine transferase enzyme encoded by genes *pmrC* (*eptA*) or *mcr*. The two component systems PmrA/B and PhoP/Q control the expression of the gene encoding the enzyme. Other mechanisms of acquiring resistance include abrogation of lipid A synthesis due to mutations

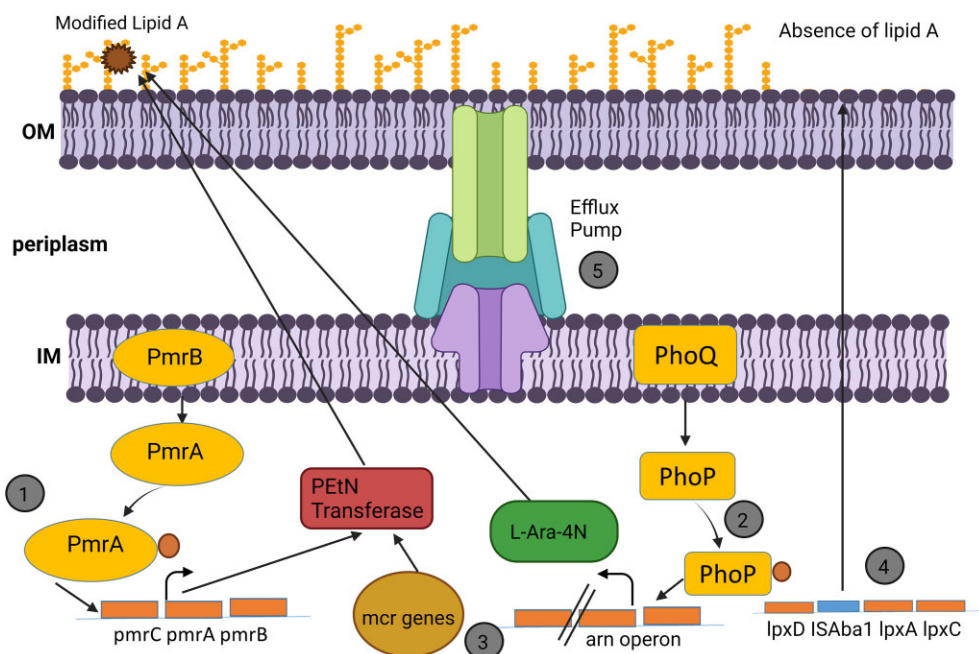


Figure 1. Mechanisms of colistin resistance. (1) Sensor kinase PmrB gets activated and in turn phosphorylates the response regulator PmrA. Phosphorylated PmrA induces the expression of *pmrABC* operon leading to expression of phosphoethanolamine transferase (PEtN transferase) and subsequent modification of lipid A. Colistin cannot bind to modified lipid A. (2) Sensor kinase PhoQ in response to cues, phosphorylates the response regulator PhoP, which induces expression of *arn* operon genes leading to addition of L-Ara4N to lipid A. (3) *mcr* genes encode the phosphoethanolamine transferase gene and lead to the addition of PEtN to lipid A. (4) In *A. baumannii*, insertional inactivation of lipid synthesis genes *lpx* DAC leads to absence of lipid A on the LPS. (5) Overexpression of efflux pump genes leads to efflux of the antibiotic from the cell. IM—inner membrane; OM—outer membrane. Image created with BioRender.com.

or inactivation of lipid synthetic genes *lpxA*, C, or D and overexpression of efflux pumps (Novovic and Jovicic 2023). Colistin resistance can be chromosomal or plasmid-mediated. The phosphoethanolamine transferase encoding *mcr* genes can be horizontally transferred making them important in community transmission of colistin resistance. Since the first report of *mcr*-1 gene in 2015, 10 variants of *mcr* (1–10) have been reported (Cherak et al. 2021). Different variants are reported from bacteria originating from different sources, including humans, animals, vegetables, water, etc. with *mcr*-1 and *mcr*-9 being the most disseminated variants (Martiny et al. 2022). Though *mcr*-1 and *mcr*-9 were reported to be most abundant among human metagenome samples, the environmental origins of other variants showed more variability (Martiny et al. 2022). Most importantly, *mcr* gene carrying plasmids co-harbours other resistance genes, including carbapenemase and extended-spectrum beta-lactamase (ESBL) genes, making them important from surveillance point of view.

Recent developments on detection of colistin resistance—phenotypic and molecular assays

Early and accurate colistin resistance detection is vital for AMR surveillance and controlling community spread. These methods include phenotypic and molecular detection (Fig. 2). While phenotypic tests identify resistance using solid/liquid media, molecular methods pinpoint genetic resistance mechanisms like mutations. Colistin resistance involves multiple genes, necessitating combined genetic and phenotypic analysis. Since the previous reviews (Bardet et al. 2019, Sekyere 2019, Leshaba et al. 2022), many recent advancements have

marked the field. Here, we explore new detection methods alongside the established ones.

Phenotypic methods

Phenotypic methods require isolating pure bacterial culture from the clinical/environmental samples before performing the assays. Although both agar- and broth-based measurements of colistin resistance are available, CLSI and EUCAST guidelines do not recommend disc diffusion methods for colistin, as the large size of colistin does not ensure optimum diffusion in agar. However, CLSI approved the use of colistin broth disc elution (CBDE) and colistin agar test (CAT) methods for susceptibility testing, although broth microdilution remains the reference method. These tests are still used in resource-limited settings (Osei Sekyere et al. 2020). On the contrary, EUCAST accepts the BMD alone as the acceptable method for colistin resistance determination. Phenotypic methods have been already extensively reviewed (Sekyere 2019, Leshaba et al. 2022) and a brief overview of different phenotypic tests is discussed below and their merits and disadvantages summarized in Table 1.

Broth-based tests

BMD method

This is currently the gold standard method for colistin resistance determination recommended jointly by the CLSI and EUCAST, and involves the preparation of a series of two-fold dilutions of colistin sulfate in polystyrene plates in cation-adjusted Mueller Hinton broth without any polysorbates, followed by addition of bacterial inoculum and incu-

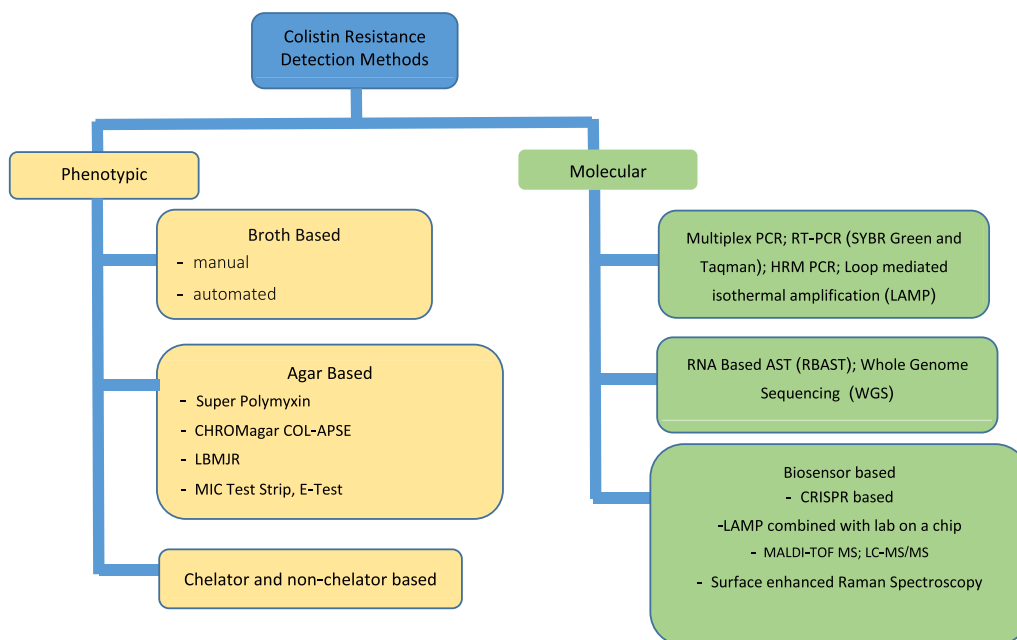


Figure 2. General classification of colistin resistance detection mechanisms. The figure gives a brief overview of the main colistin resistance methods.

bation at 37°C for 18 hours. The least concentration where no growth of the bacteria is observed, is determined as the MIC. EUCAST recommends the breakpoint for colistin at $\leq 2 \text{ mg l}^{-1}$ (susceptible) and $> 2 \text{ mg l}^{-1}$ (resistant), whereas CLSI has removed the susceptible category (Satlin et al. 2020).

However, BMD is not routinely practiced by all labs since it involves preparation of different dilutions of antibiotics, and the problem of colistin binding to the plastic surfaces of the microtitre wells (Karvanen et al. 2017). Moreover, ‘skipped well’ phenomenon has been observed with colistin BMD with certain bacteria like *Acinetobacter* and *Pseudomonas* (Elias et al. 2021).

The CLSI has recommended the use of CBDE test and CAT for *Enterobacterales* and *P. aeruginosa* based on their performance in a multicentre evaluation study (Humphries et al. 2019). The CBDE uses commercially available colistin discs ($1\text{--}4 \mu\text{g ml}^{-1}$), which are added to predefined Mueller Hinton broth. CAT includes colistin in Mueller Hinton agar and documenting the presence or absence of bacterial growth (Humphries et al. 2019). Overall, both tests showed around 98% Categorical Agreement, 94% (CBDE) and 96% (CAT) Essential Agreement, 3.2% (CBDE) and 3.9% (CAT) Very Major Error and almost nil Major Error (Humphries et al. 2019).

Non-automated and automated broth-based tests

These are modifications of the BMD method. Non-automated methods, such as the UMIC test kit, MICRONAUT MIC Strip, and Com ASP Colistin, come with strips preloaded with dehydrated colistin in a range of concentrations. Since they come with preloaded antibiotic, they are easier to perform compared to the classical BMD. Automated methods include commercial MIC determination systems based on the broth microdilution and provide simultaneous identification of the bacterial species. These methods are reported to be more concurrent with the gold standard method and are preferred for

their ease of use. Automated broth-based tests include the Microscan, MICRONAUT S, BD Phoenix, Sensititre, and Vitek 2 (Table 1). When compared with the BMD method, most of these tests had acceptable EA and CA values. However, methods like the Microscan and Vitek 2 had high VME (10% and 36%) compared to BMD (Chew et al. 2017, Lutgring et al. 2019).

Agar-based tests

These tests rely on measuring the zone of inhibition formed by the diffused antibiotic or a dye-based colour change, as a result of a change in pH due to bacterial growth in an agar-based medium containing the indicator bacteria. Details of different agar-based methods are given in Table 1. While SuperPolymyxin medium uses methylene blue dye-based agar with daptomycin and amphotericin B, LBJMR medium uses purple agar base with vancomycin (Nordmann et al. 2016b, Bardet et al. 2017). However, agar dilution test has been approved by CLSI in 2020 for colistin resistance detection, while EUCAST accepts only the BMD method. The method, however, is labour intensive in terms of the several agar plates to be prepared for the different concentration ranges and replicates.

Other methods

Chelator-based tests

Chelator-based tests can differentiate colistin-resistant, *mcr*-positive/negative isolates. These tests are based on *mcr*-encoded phosphoethanolamine transferase, a zinc-dependent enzyme. Chelation of zinc abrogates the enzyme function and reverses resistance. Thus, colistin resistant non-*mcr* isolates will grow on the plates, whereas *mcr*-positive, resistant isolates will not grow. Using this principle, colistin resistant non-*mcr* strains of *E. coli* were distinguished based on the growth on colistin-EDTA (ethylenediamine tetraacetic acid) plates (Gonzales Escalante et al. 2020). Colistin-

Table 1. Phenotypic, molecular, and Omics-based methods of screening colistin resistance.

Method	Principle	Advantage	Limitations	Agreement with BMD*	Turnaround time (in hours)	Reference
Phenotypic methods of detection						
Broth microdilution (BMD)	Detection of presence or absence of growth of bacteria in broth containing serial dilutions of colistin based on absorbance measurements	Gold standard method recommended by CLSI and EUCAST; Several concentrations and isolates can be tested simultaneously	Variability with some strains like <i>Acinetobacter</i> , <i>Pseudomonas</i> , etc.; Skipped well phenomenon; Adherence of colistin to plastic surfaces; labour intensive	Not applicable	24	(ECDC 2019)
UMIC and MICRONAUT MIC Strip (MMS)	Broth based non-automated; different dilutions of colistin	Readymade strips with different concentrations of antibiotic, no need for manual dilutions	Could not detect resistance in <i>Acinetobacter</i> , <i>Pseudomonas</i> , etc.	EA—94%; 82% CA—100%; 92% ME—nil; 4% VME—nil; 4%	18–24; 18–24	(Bardet et al. 2019), (Matuschek et al. 2018)
ComASP Colistin/Sensitest	BMD based; Turbidity based detection of growth	No need for manual dilutions; Single panel can assay four isolates	Only 7 dilutions as against 11 dilutions in UMIC/MMS; Cannot detect resistance in all bacterial species	EA—88% CA—89% ME—9% VME—1%	16–20	(Matuschek et al. 2018)
Microscan	Based on BMD principle; Fluorimetric detection of growth	Automated, easy to perform	Most expensive of several tests compared; Overestimated MIC of non-fermentative GNB	CA 88%; VME of 4%; ME 15.8%; CA 91.9%; overall ME 26.9%; CA 92.7%; VME 10.7%.	18–24	(Chew et al. 2017, Jayol et al. 2018, Lutgring et al. 2019)
MICRONAUT S	Photometric detection	Automated; Multiple isolates can be tested in a single plate	False susceptible results (VME), especially for <i>Pseudomonas</i>	EA 96%; CA 89%; ME 9.3%; VME 2.6%	18–24	(Matuschek et al. 2018)
BD Phoenix	Biochemical identification of isolates and fluorimetric detection of susceptibility	Automated; combines bacterial identification with susceptibility testing reducing turnaround time	Showed high rate of false susceptibility results	VME 12%	18–24	(Jayol et al. 2018)
Vitek 2		Semi-automated; 96.1% CA with BMD		VME of 36%	18–24	(Chew et al. 2017)
SuperPolymyxin	Selective medium consisting of eosin-methylene blue dye along with colistin, daptomycin and amphotericin B	Sensitivity 95.2%; Specificity 95.3% using cultured bacteria; could detect <i>mcr-1</i> positive isolates efficiently	High number of colistin susceptible isolates grew on medium; higher workload in terms of cross validation	CA 82.1%; ME 7.4%; VME 10.2%	24–48	(Przybysz et al. 2018), (Nordmann et al. 2016a)
LBJMR	Selective medium with purple agar base containing colistin and vancomycin	Can detect colistin resistant Enterobacteriaceae and vancomycin resistant Enterococci; <i>mcr-1</i> carrying isolates with low colistin MIC were also picked up	Lack of many studies evaluating with other methods	100% sensitivity and specificity	18–24	(Bardet et al. 2017)
CHROMagar COL-APSE	Chromogenic selective medium containing colistin and oxazolidinones	Can identify resistant non-fermentative Gram-negative bacteria; easy to perform	Comparison with MicroScan, Com-ASP and colistin-MAC test revealed Chromagar to be the least sensitive and specific	CA 73.81%; ME 33.33%; VME 17.95%;	24	(Osei Sekyere et al. 2020)

Table 1. Continued

Method	Principle	Advantage	Limitations	Agreement with BMD*	Turnaround time (in hours)	Reference
E-test (Epsilonometer Test) and MIC Test Strip (MTS)	Strip containing continuous, exponential gradient of colistin along the length of the strip	Easy to perform since predefined colistin concentrations are present on the strip	Poor diffusion of colistin on the agar leading to error in MIC judgement (high VME)	EA 75%; CA 92%; ME 5.9%; VME 12.%;	24; 16–20	Chew et al. 2017 , (Hindler and Humphries 2013)
Polymyxin NP	Glucose-based medium containing colistin. As bacteria grow, glucose is utilized and medium turns acidic which is detected based on colour change of dyes (phenol red)	Good sensitivity (100%) and specificity	Cannot be used for non-fermentative Gram-negative bacteria	VME 1.2%	<2	Jayol et al. 2018
Modified Rapid Polymyxin Test	Chelator based; uses EDTA	Can identify <i>mcr</i> positive resistant isolates based on reversal of colistin susceptibility	Lower specificity for <i>K. pneumoniae</i>	Not available	<2	(Esposito et al. 2017), (Leshaba et al. 2022)
Molecular methods of detection						
Conventional PCR	Amplification of specific gene of interest	Can process several samples simultaneously; multiplex PCR can identify several <i>mcr</i> genes in a single reaction	Cannot determine MIC of antibiotic; cannot determine resistance conferring mutations	Not applicable	<3	(Rebelo et al. 2018)
Real-time PCR	Uses fluorescent dyes that emit fluorescence as amplicons form	Real-time visualization of data; abrogating the need for downstream processing	Cannot determine MIC of antibiotic; cannot determine resistance conferring mutations	Not applicable	<3	(Li et al. 2017 , Tolosi et al. 2020 , Mentasti et al. 2021)
High-resolution melt (HRM) curve analysis	Careful denaturation of double stranded amplicon post real-time PCR and its melt curve analysis; change in base pair may lead to shift in curve	Can be coupled with real-time PCR and mutations can be analysed	Requires dedicated software for accurate analysis	Not applicable	3–4	(Tahmasebi et al. 2020a, b, c)
Loop-mediated isothermal amplification (LAMP)	Set of 4–6 primers representing different regions of a gene form looped product. Strand displacement property of <i>Bst</i> polymerase displaces the strand forming a new extension cycle and amplification from looped strand can be initiated by the polymerase	Rapid amplification reaction at isothermal condition, no need for thermal cyclers; highly sensitive; products can be visualized by different ways.	Primer design requires expertise; not easily amenable to multiplex PCR reactions	Not applicable	<1	(Notomi et al. 2000 , Zhong et al. 2019 , Rodriguez-Manzano et al. 2020 , Gong et al. 2022)

Table 1. Continued

Method	Principle	Advantage	Limitations	Agreement with BMD*	Turnaround time (in hours)	Reference
Recombinase polymerase amplification (RPA) combined with CRISPR/Cas12a	A recombinase primer complex identifies and binds to cognate region in target DNA and a strand displacing polymerase elongates the primer. CRISPR/Cas12a system was used to produce characteristic cleaved bands	Low isothermal operating temperatures (22°C–45°C), hence no need for thermal cyclers; high selectivity and specificity; rapid amplification of products; amenable to multiplexing	Not efficient in amplifying larger DNA fragments	Not applicable	<1	(Lobato and O'Sullivan 2018, Gong et al. 2022)
Multiple cross displacement amplification (MCDA) coupled with lateral flow device	Similar to LAMP, but uses random primers; technique coupled to a gold nanoparticle streptavidin and biotinylated antibody based lateral flow device	Highly sensitive; could detect 600 fg of <i>mcr-1</i> -positive plasmid DNA per microliter	Only one sample and gene could be tested per assay	Not applicable	<1	(Gong et al. 2019)
Omics-based detection methods (Genomics/Transcriptomics/Proteomics/Metabolomics)						
Whole-genome sequencing (WGS)	Next generation sequencing using Illumina or Minion platform	Comprehensive knowledge of genes and mutations and identification of plasmid-borne AMR genes	Prohibitory cost compared to existing methods; higher skill and expertise needed	Not Applicable	<48	(Schurch and van Schaik 2017, Ragupathi 2019, Khedher et al. 2020)
RNA-based antimicrobial susceptibility testing (RBAST)	Differential expression of certain mRNA transcripts (RNA biomarkers) in <i>mcr-1</i> positive resistant isolates when compared with colistin susceptible isolates	(1) Rapid compared to conventional methods (2) Simultaneously detect multiple resistance markers or genes in a single test (3) Clinical samples, such as blood or urine can be directly analysed without the need for lengthy culture steps	(1) May fail to accurately determine susceptibility due to genetic variability in bacterial population (2) Not cost effective (3) Host RNA may potentially interfere with the accuracy of RBAST assay	Not Applicable	3	(Zhang et al. 2022)
MALDI-TOF MS	Biochemical changes in lipid A in resistant bacteria are detected by ionizing the samples by MALDI-TOF MS. The change in m/z peaks can differentiate resistant and susceptible bacteria	Direct analysis from patient samples possible; can determine the biochemical changes undergone	Prohibitive cost in terms of equipment; needs highly skilled personnel for evaluation of results	Not applicable	0.5–1	(Dortet et al. 2018, Furniss et al. 2019, Smith et al. 2021, Smith et al. 2022)

Table 1. Continued

Method	Principle	Advantage	Limitations	Agreement with BMD*	Turnaround time (in hours)	Reference
Liquid chromatography-mass spectrometry (LC-MS/MS)	Combination of liquid chromatography(LC) and mass spectrometry (MS), which enables the separation, identification, and quantification of molecules present in a biological sample.	(1) High sensitivity; (2) Multiple resistance markers can be analysed; (3) Possible to quantify the target molecule; (4) Compatible with various samples types including clinical, environmental, and microbial cultures.	(1) LC-MS can detect only known resistance markers for colistin resistance, and may detect novel or undiscovered resistance mechanisms. (2) Need Standardization due to limited number of studies (3) Requires specialized equipment and skilled personnel. (4) Time-consuming. LC-MS analysis is generally performed on a sample-by-sample basis, and screening a large number of samples for colistin resistance require more time.	Not applicable	<1.5	(Wang et al. 2019)
Surface- enhanced Raman spectroscopy (SERS)	Identifies molecular signature spectra of resistant bacteria due to biochemical changes by analysing their Raman spectra. The Raman spectra are enhanced by use of nanoparticles	Avoids lengthy sample preparation procedures	Prohibitory cost of equipment and requirement for highly skilled personnel	Not applicable	<2	(Wei et al. 2018 , Weiss et al. 2019 , Ciloglu et al. 2022 , Mushtaq et al. 2022 , Saikia et al. 2022)

*The CLSI guidelines require that all phenotypic tests be compared against the BMD method. Comparison of methods should be based on the following: Essential Agreement (EA)—occurs when an MIC result is within two-fold dilution from the BMD's MIC. Essential agreement should be $\geq 90\%$ for a reliable test. Categorical Agreement (CA)—refers to agreement in the interpretation of the MIC between the test and BMD. The CA should be $\geq 90\%$. Major Error (ME)—occurs when the tested MIC is resistant while BMD MIC is susceptible (false positive). The ME should be $\leq 3\%$. Very major error (VME)—occurs when the evaluated method's MIC is susceptible while BMD MIC is resistant (false negative). The VME should be $\leq 1.5\%$.

MAC test uses dipicolinic acid as chelator (Coppi et al. 2018).

The Modified Rapid Polymyxin NP (MRPNP) test was modified by adding EDTA to the medium with and without colistin. Wells with colistin and EDTA inhibited the growth of *mcr*-mediated colistin-resistant isolates due to chelation of zinc, reducing their MIC (Esposito et al. 2017). Chelator-based tests have proven to be convenient, reliable, and they can be used to detect the presence of *mcr* genes.

Molecular methods

Phenotypic tests fail to determine MIC of *mcr*-mediated resistance efficiently, as some *mcr*-gene carriers remain susceptible. This results in error near the breakpoint concentration (Chew et al. 2017). The skip-well phenomenon further complicates MIC interpretation. Precise identification of plasmid-mediated *mcr* genes demands advanced molecular methods. These methods pinpoint genetic mechanism associated with the resistance and reduces false negatives. Recent progress in molecular detection of colistin and *mcr* is summarized below.

Polymerase Chain Reaction (PCR) -based assays

Conventional and real-time PCR detect the presence of *mcr* genes rapidly. Multiplex PCR allows rapid detection of multiple *mcr* gene variants simultaneously from large samples (Rebello et al. 2018). Multiplex real-time PCR shortens resistance detection time with real-time visualization. The method successfully detected *mcr* genes (1–3) in *E. coli* at a lower limit of 10^2 bacteria (Li et al. 2017). In another study, *mcr* genes 1–5 were detected in bacterial isolates and environmental samples (Tolosi et al. 2020).

High-resolution melt (HRM) curve analysis

The process involves melting of PCR amplicons and analysing melt curves with dedicated HRM software. A screening study involving 101 *P. aeruginosa* isolates for carbapenem/colistin resistance identified four *pmrA* mutations, three *mcr*-1, and various *bla* variants (Tahmasebi et al. 2020b). Similarly, co-existing *mbi*, *ampC*, *mcr*-1 genes were identified from bloodstream infection-associated *P. aeruginosa* via HRM (Tahmasebi et al. 2020c). The method detects *mcr*-1 from plasmid DNA at 10^3 CFU/ml, showing high sensitivity and specificity (Tahmasebi et al. 2020a).

Loop-mediated isothermal amplification (LAMP)

The LAMP assay is a simple, reliable, rapid, and sensitive method for detection of genes of interest. This technique performs amplification at constant temperature and does not require sophisticated PCR machines (Notomi et al. 2000). The amplicons obtained can be visualized by colorimetry, fluorimetry, agarose gel electrophoresis, or even a lateral flow assay (Park 2022). LAMP has been adopted to detect *mcr* genes with different readout formats to improve the efficiency and ease of detection. A LAMP-based Lab-on-a-Chip device for detecting *mcr*-9 gene from clinical isolates was developed (Rodriguez-Manzano et al. 2020). The system combined LAMP with a semiconductor technology to detect amplicons and detected *mcr*-9 gene in 10 minutes and showed 100% concordance with the WGS data (Rodriguez-Manzano et al. 2020). The only time-limiting steps were obtaining the

pure bacterial culture and DNA isolation. The major disadvantage with this system is that only one gene and sample could be analysed at a time (Rodriguez-Manzano et al. 2020).

In another study, multiplex LAMP-based detection of *mcr* 1–5 genes was achieved by using multiple sets of primers corresponding to each *mcr* gene (Zhong et al. 2019). The primers were modified by addition of *Hind* restriction site. The amplicons were digested with the enzyme and the specific pattern of the digested fragments was used to identify the *mcr* gene of interest (Zhong et al. 2019).

Recently, a recombinase polymerase amplification (RPA) assay combined with clustered regularly interspaced short palindromic repeats (CRISPR)/Cas12a technology was developed to detect *mcr*-1 gene (Gong et al. 2022). The system used a recombinase enzyme–primer complex to identify a homologous sequence on the target *mcr*-1 gene, which contained a protospacer adjacent motif (PAM) site. Isothermal amplification at 37°C mitigated the need for thermocyclers. Detection of amplicons was facilitated by recognition of the PAM site by the CRISPR and activation of Cas12a, which cleaves the fluorescent labelled probes resulting in emission of fluorescence (Gong et al. 2022; Fig. 3). The whole assay could be completed in 1 hour and was sensitive (Gong et al. 2022). The RPA amplification is amenable to multiplexing, and hence can be used to screen multiple variants of the *mcr*. Moreover, different methods like agarose gel electrophoresis, colorimetry, fluorimetry, and spectroscopy can be adapted to detect the amplified products (Lobato and O' Sullivan 2018).

In another study, multiple cross displacement amplification (MCDA) was coupled to a gold nanoparticle-based lateral flow device to detect the *mcr*-1 gene (Gong et al. 2019). The method was highly sensitive as it was able to detect 600 fg of *mcr*-1-positive plasmid DNA per microlitre in pure culture and 4.5×10^3 CFU ml⁻¹ in fecal samples spiked with 100 µl of strains. However, only one sample and gene can be tested per sensor slide (Gong et al. 2019).

Whole-genome sequencing

Next-generation sequencing technologies have opened up new possibilities in AMR detection and analysis. The accessibility of whole-genome sequences of large number of bacterial isolates has facilitated retrospective analysis of the genomes and identification of otherwise undetected antibiotic resistance mutations.

WGS has been used as part of surveillance for the presence of plasmid-mediated resistance genes. Long-read platforms cover up to 60 kb regions, including repetitive regions, thereby facilitating easier assembly of whole genomes (Schurch and van Schaik 2017). The Comprehensive Antibiotic Resistance Database (CARD; <https://card.mcmaster.ca/>) and ResFinder are curated databases that maintain information on annotated antibiotic resistance genes, their ontology, sequence variants, and horizontal transfer (Florensa et al. 2022). Analysing the assembled sequence data against these databases helps in identifying the antibiotic resistance mutations to the level of single nucleotide variation (Devanga Ragupathi et al. 2020). Plasmid replicons can be identified from the WGS data with the help of software like PlasmidFinder (Ragupathi NKD et al. 2019, Devanga Ragupathi et al. 2020; Azam M 2021). The *mcr*-9 gene is inducible upon colistin exposure if it is located

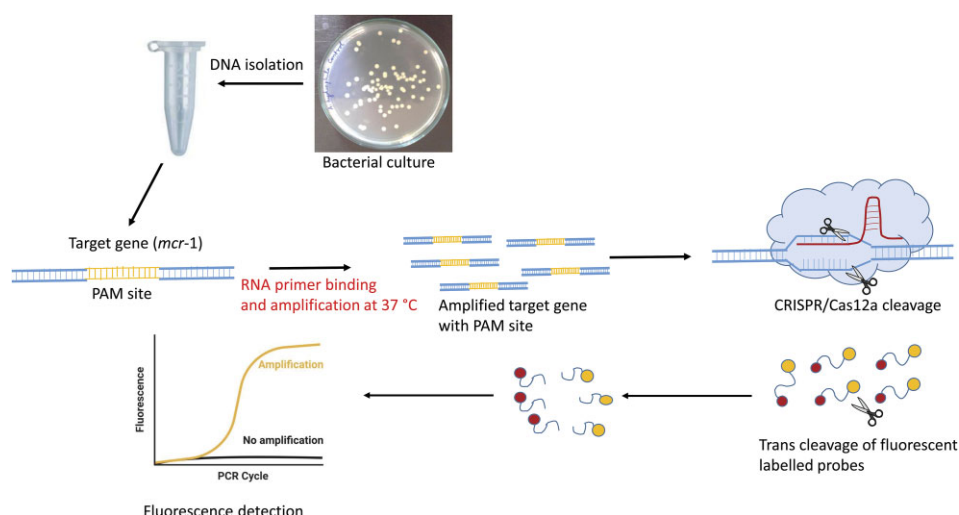


Figure 3. Recombinase polymerase amplification and CRISPR Cas12a-based detection of *mcr-1* gene.

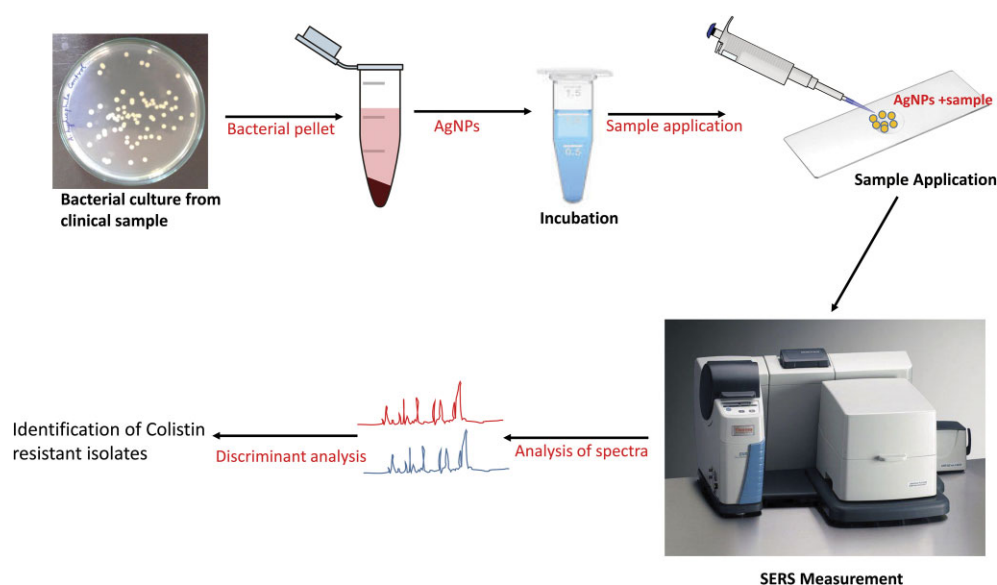


Figure 4. Surface-enhanced Raman spectroscopy based detection of colistin resistance from bacterial cultures.

downstream of the genes *qseBC* in a plasmid (Kieffer et al. 2019). Also, *mcr* genes often co-occur with other resistance genes like the carbapenemase genes. Thus, identification of the genetic background of a gene is important to know whether the gene is induced/silenced or co-harbours other resistance genes. Some groups have identified the genetic background of *mcr-9* gene in the respective plasmid by means of WGS, revealing their expression status (Ragupathi 2019; Ragupathi et al. 2020, Singh et al. 2021). In a retrospective bioinformatics analysis of 64 628 Gram-negative bacterial genomes from the NCBI database, a total of 1386 *mcr* hits were identified from different bacteria (Khedher et al. 2020). The *mcr* variants identified included *mcr-1*, 2, 3, 4, 5, 8, and 9. It also revealed that *mcr-9* was the only gene identified among isolates obtained exclusively from human sources (Khedher et al. 2020).

RNA-based Antimicrobial Susceptibility Testing (RBAST)

The method detects resistance by tracking the mRNA changes induced in bacterial cells upon exposure to colistin. Differentially expressed genes of colistin-susceptible (DH5 α -pUC19) and -resistant (DH5 α -pUC19 -*mcr-1*) *E. coli* in response to 1 hour treatment of breakpoint concentration of colistin was compared by RNA sequencing (Zhang et al. 2022). After analysing all the differentially expressed gene ontologies, the authors identified a set of 12 genes as RNA biomarkers that were highly overexpressed in colistin-susceptible isolates compared to *mcr-1* carrying colistin-resistant isolates. These RNA biomarkers could be used for rapid detection of *mcr*-mediated colistin resistance. The method was validated with 30 clinical strains and was correlated with other AST tests and the accuracy was found to be >93% (Zhang et al. 2022).

Biosensor-based colistin resistance detection methods

Electrochemical impedance spectroscopy biosensors

Electrochemical-based biosensors have been devised for the detection of multidrug-resistant bacteria. They use probes based on specific enzymes or nanoparticles to detect the electrochemical signals.

In a study, a specific isothermal amplification system called toehold-mediated strand displacement was used to amplify and detect the presence of *mcr-1* gene (Li et al. 2018). The group combined this method to a methylene blue-based detection of the amplicons. It makes use of the fact that methylene blue (MB) can intercalate into double stranded DNA (dsDNA) causing decreased diffusion of the MB/dsDNA complex, thereby distinguishing single stranded and double stranded DNA (Li et al. 2018). The specific hairpin probe was designed for *mcr-1* and it was combined on an electrochemical platform with different electrodes, which were connected to a sensor. The output was measured as volts (Li et al. 2018).

Matrix Assisted Laser Desorption Ionization–Time Of Flight Mass Spectrometry (MALDI-TOF MS)

The MALDI-TOF-based AMR detection identifies differences in the MS peaks of biochemical metabolites of resistant bacteria as a result of exposure to antibiotics. In case of colistin resistance mediated by *mcr* genes, the addition of phosphoethanolamine moiety to lipid A has been detected by MALDI-TOF mass spectrometry (Furniss et al. 2019, Smith et al. 2022). A comparative MALDI-TOF MS analysis of colistin-resistant and -sensitive *A. baumannii* was performed by one study (Dortet et al. 2018). The presence of phosphoethanolamine group on the lipid A of resistant strains resulted in +25 and +123 shifts of mass unit of the native phosphorylated hepta-acyl lipid A molecule giving rise to dominant peaks at *m/z* 1935.3 and *m/z* 2033.3 for colistin-resistant strains as against peaks at *m/z* 1728.1 and *m/z* 1910.3 for sensitive strains. However, the negative ion mode used for the ionization in this study may not be commonly available in the machines used routinely in diagnostic labs (Dortet et al. 2018).

A fast lipid analysis technology applied with MALDI-TOF detected colistin resistance in bacteria directly from patient urine samples. A mere 1 µl of urine was applied to a MALDI plate, then heated at 110°C for 30 minutes for lipid extraction. After washing, matrix overlay and MALDI-TOF analysis followed. Resistant samples exhibited peaks at *m/z* 1797 and *m/z* 1920, consistent with *E. coli* lipid A's phosphoethanolamine structure. PCR analysis confirmed the presence of *mcr* genes, correlating with spectral peaks. The total time for the assay was under 1 hour (Smith et al. 2021).

In a study using MALDI TOF Biotyper Sirius MS, researchers divided the sum of intensities of peaks for phosphoethanolamine (PeTn) or L-4-aminoarabinose (L-Ara4N) by the intensity of the peak for native lipid A, defining a polymyxin resistance ratio (PRR; Furniss et al. 2019). The study predicted the colistin resistance mechanism by assessing PRR values for PeTn (PRR1919) or L-Ara4N (PRR 1927). Validation with various *E. coli* strains revealed that isolates with PRR1919 value equal to the PRR values likely carried *mcr* genes, while those with PRR values equal to

PRR1927 had chromosomal mutations like *pmrAB* (Furniss et al. 2019). MALDI-TOF avoids lengthy sample processing time, but demands expertise in chemistry and spectroscopic techniques.

Liquid chromatography–mass spectrometry (LC–MS/MS)

A genoproteomic approach was used to identify *mcr-1*-mediated colistin resistance using LC–MS/MS approach (Wang et al. 2019). Tryptic peptides of MCR-1 protein were compared to protein sequences of several mutant variants of *mcr-1* and a set of core tryptic peptides common to all variants were identified. These were compared with other *mcr* genes (2–4) and only those that were unique to *mcr-1* was selected. Three core peptides were thus identified. The method was validated using proteins from *mcr-1*-positive isolates and -negative isolates (Wang et al. 2019). An isolate was concluded as MCR-1 positive if its tryptic peptide MS/MS fragments had at least two of the three core peptides. The assay was validated using blinded 99 clinical isolates containing *mcr-1* gene and by three independent blinded operators. The method did not involve cumbersome protein preparation processes, but required a highly sensitive LC–MS system (Wang et al. 2019).

Surface-enhanced Raman spectroscopy (SERS)

Raman spectroscopy involves measuring the Raman scatter produced by samples upon induction by a light source (photons). Since the natural Raman scatter by samples can be enhanced by employing nanoparticles as substrates, Raman spectroscopy does not require any labelling and unlike infrared spectroscopy, is not affected by water (Saikia et al. 2022). SERS identifies molecular signature spectra of bacteria arising due to the differences in biochemical changes (Wei et al. 2018; Fig. 4). The difference in lipid A moiety between colistin-susceptible and -resistant strains of bacteria can be detected by utilizing the surface-enhanced Raman spectroscopy (Mushtaq et al. 2022, Saikia et al. 2022). By using a silver nanoparticle enhanced Raman spectroscopy, Saikia et al., showed that a distinct peak at 757 nm differentiates colistin-resistant and -sensitive *E. coli* (Saikia et al. 2022). One group used a machine learning-based feature extraction by autoencoder (non-linear feature extraction) and principal component analysis (linear feature extraction) and a support vector machine to identify the distinguishing spectral details (Ciloglu et al. 2022). They found that the non-linear feature extraction using the autoencoder was able to efficiently identify the spectral features that separated colistin-resistant *K. pneumoniae* with sensitive *K. pneumoniae* (Ciloglu et al. 2022). The method did not require elaborate sample preparation procedures and they simply mixed the bacterial sample with the prepared silver nanoparticles and obtained a high signal-to-noise ratio (Ciloglu et al. 2022).

Discussion

Recent reports of *mcr*-mediated colistin resistance underlines the necessity for constant AMR surveillance. At the same time, colistin susceptibility testing comes with many challenges. The reference standard method BMD, is not feasible in many resource constrained laboratories. Recently, CLSI approved the

use of CBDE test and CAT for resistance detection. Given their ease of testing, these tests could be of immense help in resource-limited settings. Molecular methods reduce the turnaround time of resistance detection and provide much more information than phenotypic methods. Considering the urgency and accuracy of the tests, a combination of two tests may prove highly effective. An automated broth microdilution test combined with a molecular test, like multiplex real-time PCR or LC-MS/MS may aid in the rapid identification of the bacterial species, determining the MIC as well as identifying the resistance gene. Rapid detection of resistance is crucial for timely treatment decisions. WGS may be useful in further characterization of resistant strains, but the cost remains prohibitory to many labs.

Conclusion

The major limitations of the current colistin resistance detection lie in the accuracy of determining *mcr*-based resistance and cost-effectiveness. However, some molecular tests like mass spectrometry or SERS, even though accurate in detecting *mcr*-mediated resistance, may be out of reach for many clinical settings. In such scenario, these techniques could be used by research laboratories in aiding clinicians in plasmid-mediated resistance surveillance at regular intervals. An understanding of the prevalence of plasmid-mediated resistance genes among environmental/clinical isolates can help in policy decisions to prevent AMR spread.

Acknowledgements

We thank Dr Agiesh Kumar, Associate Professor, MGMARI, Sri Balaji Vidyapeeth for his valuable suggestions.

Conflict of interest

The authors declare no conflict of interest.

Funding

We acknowledge Sri Balaji Vidyapeeth for providing facility and financial support (intramural seed funding to DL-SBV/IRC/SEED MONEY/138/2023; DR—SBV/IRC/SEED MONEY/145/2023; VS—SBV/IRC/SEED MONEY/135/2022).

Author contributions

Divya Lakshmanan (Conceptualization [lead], Data curation [lead], Formal analysis [lead], Writing – original draft [lead], Writing – review & editing [lead]), Dhamodharan Ramasamy (Data curation [equal], Formal analysis [equal], Writing – original draft [equal], Writing – review & editing [equal]), Veni Subramanyam (Formal analysis [supporting], Writing – original draft [supporting], Writing – review & editing [supporting]), and Suresh Kumar S (Data curation [supporting], Writing – original draft [supporting], Writing – review & editing [supporting])

Data availability

No new data were generated or analysed in support of this research.

References

- Azam M, Gaid R, Yadav G *et al.* Colistin resistance among multiple sequence types of *Klebsiella pneumoniae* is associated with diverse resistance mechanisms: a report from India. *Front Microbiol* 2021;12. <https://doi.org/10.3389/fmicb.2021.609840>
- Bardet L, Le Page S, Leangapichart T *et al.* LBJMR medium: a new polyvalent culture medium for isolating and selecting vancomycin and colistin-resistant bacteria. *BMC Microbiol* 2017;17:220. <https://doi.org/10.1186/s12866-017-1128-x>.
- Bardet L, Okdah L, Le Page S *et al.* Comparative evaluation of the UMIC Colistine kit to assess MIC of colistin of Gram-negative rods. *BMC Microbiol* 2019;19:60. <https://doi.org/10.1186/s12866-019-1424-8>.
- Cherak Z, Loucif L, Moussi A *et al.* Epidemiology of mobile colistin resistance (*mcr*) genes in aquatic environments. *J Glob Antimicrob Resist* 2021;27:51–62. <https://doi.org/10.1016/j.jgar.2021.07.021>.
- Chew KL, La MV, Lin RTP *et al.* Colistin and Polymyxin B susceptibility testing for carbapenem-resistant and *mcr*-positive Enterobacteriaceae: comparison of Sensititre, MicroScan, Vitek 2, and Etest with broth microdilution. *J Clin Microbiol* 2017;55:2609–16. <https://doi.org/10.1128/JCM.00268-17>.
- Ciloglu FU, Hora M, Gundogdu A *et al.* SERS-based sensor with a machine learning based effective feature extraction technique for fast detection of colistin-resistant *Klebsiella pneumoniae*. *Anal Chim Acta* 2022;1221:340094. <https://doi.org/10.1016/j.aca.2022.340094>.
- Coppi M, Cannatelli A, Antonelli An *et al.* A simple phenotypic method for screening of MCR-1-mediated colistin resistance. 2018, , 24, 201.e1–201
- Devanga Ragupathi NK, Veeraraghavan B, Muthurulandi Sethuvel DP *et al.* First Indian report on genome-wide comparison of multidrug-resistant *Escherichia coli* from blood stream infections. *PLoS One* 2020;15:e0220428. <https://doi.org/10.1371/journal.pone.0220428>.
- Dortet L, Potron A, Bonnin RA *et al.* Rapid detection of colistin resistance in *Acinetobacter baumannii* using MALDI-TOF-based lipidomics on intact bacteria. *Sci Rep* 2018;8:16910. <https://doi.org/10.1038/s41598-018-35041-y>.
- ECDC. *Laboratory Manual for Carbapenem and Colistin Resistance Detection and Characterisation for the Survey of Carbapenem-and/or Colistin-resistant Enterobacteriaceae*. Stockholm: European Centres for Diseases Control, 2019.
- Elias R, Melo-Cristino J, Lito L *et al.* *Klebsiella pneumoniae* and colistin susceptibility testing: performance evaluation for broth microdilution, agar dilution and minimum inhibitory concentration test strips and impact of the “skipped well” phenomenon. *Diagnostics (Basel)* 2021;11:2352.
- Esposito F, Fernandes MR, Lopes R *et al.* Detection of colistin-resistant MCR-1-positive *Escherichia coli* by use of assays based on inhibition by EDTA and zeta potential. *J Clin Microbiol* 2017;55:3454–65. <https://doi.org/10.1128/JCM.00835-17>.
- Florensa AF, Kaas RS, Clausen P *et al.* ResFinder—an open online resource for identification of antimicrobial resistance genes in next-generation sequencing data and prediction of phenotypes from genotypes. *Microb Genom* 2022;8:000748.
- Furniss RCD, Dortet L, Bolland W *et al.* Detection of colistin resistance in *Escherichia coli* by use of the MALDI Biotyper Sirius mass spectrometry system. *J Clin Microbiol* 2019;57. <https://doi.org/10.1128/JCM.01427-19>.
- Gogry FA, Siddiqui MT, Sultan I *et al.* Current update on intrinsic and acquired colistin resistance mechanisms in bacteria. *Front Med (Lausanne)* 2021;8:677720. <https://doi.org/10.3389/fmed.2021.677720>.
- Gong L, Jin Z, Liu E *et al.* Highly sensitive and specific detection of mobilized colistin resistance gene *mcr*-1 by CRISPR-based platform. *Microbiol Spectr* 2022;10:e0188422. <https://doi.org/10.1128/spectrum.01884-22>.
- Gong L, Liu E, Che J *et al.* Multiple cross displacement amplification coupled with gold nanoparticles-based lateral flow biosensor for detection of the mobilized colistin resistance gene *mcr*-1. *Front Cell*

- Infect Microbiol* 2019;9:226. <https://doi.org/10.3389/fcimb.2019.00226>.
- Gonzales Escalante E, Yauri Condor K, Di Conza JA et al. Phenotypic Detection of Plasmid-Mediated Colistin Resistance in Enterobacteriaceae. 2020, , 24, e01555–19
- Hindler JA, Humphries RM. Colistin MIC variability by method for contemporary clinical isolates of multidrug-resistant Gram-negative bacilli. *J Clin Microbiol* 2013;51:1678–84. <https://doi.org/10.1128/JCM.03385-12>.
- Humphries RM, Green DA, Schuetz AN et al. Multicenter evaluation of colistin broth disk elution and colistin agar test: a report from the Clinical and Laboratory Standards Institute. *J Clin Microbiol* 2019;57:e01269–19. <https://doi.org/10.1128/JCM.01269-19>.
- Hussein NH, Al-Kadmy IMS, Taha BM et al. Mobilized colistin resistance (mcr) genes from 1 to 10: a comprehensive review. *Mol Biol Rep* 2021;48:2897–907. <https://doi.org/10.1007/s11033-021-06307-y>.
- Jayol A, Nordmann P, Andre C et al. Evaluation of three broth microdilution systems to determine colistin susceptibility of Gram-negative bacilli. *J Antimicrob Chemother* 2018;73:1272–8. <https://doi.org/10.1093/jac/dky012>.
- Karvanen M, Malmberg C, Lagerbäck P et al. Colistin is extensively lost during standard *in vitro* experimental conditions. *Antimicrob Agents Chemother* 2017;61:e00857–17. <https://doi.org/10.1128/AAC.00857-17>.
- Khedher MB, Baron SA, Riziki T et al. Massive analysis of 64,628 bacterial genomes to decipher water reservoir and origin of mobile colistin resistance genes: is there another role for these enzymes? *Sci Rep* 2020;10:5970. <https://doi.org/10.1038/s41598-020-63167-5>.
- Kieffer N, Royer G, Decousser JW et al. mcr-9, an inducible gene encoding an acquired phosphoethanolamine transferase in *Escherichia coli*, and its origin. *Antimicrob Agents Chemother* 2019;63:e00965–19.
- Leshaba TMS, Mbelle NM, Osei Sekyere J. Current and emerging polymyxin resistance diagnostics: a systematic review of established and novel detection methods. *J Appl Microbiol* 2022;132:8–30. <https://doi.org/10.1111/jam.15184>.
- Li B, Chai Z, Yan X et al. An enzyme-free homogenous electrochemical assay for sensitive detection of the plasmid-mediated colistin resistance gene mcr-1. *Anal Bioanal Chem* 2018;410:4885–93. <https://doi.org/10.1007/s00216-018-1130-7>.
- Li J, Shi X, Yin W et al. A multiplex SYBR green real-time PCR assay for the detection of three colistin resistance genes from cultured bacteria, feces, and environment samples. *Front Microbiol* 2017;8:2078. <https://doi.org/10.3389/fmicb.2017.02078>.
- Lobato IM, O'Sullivan CK. Recombinase polymerase amplification: basics, applications and recent advances. *Trends Analyt Chem* 2018;98:19–35. <https://doi.org/10.1016/j.trac.2017.10.015>.
- Lutgring JD, Kim A, Campbell D et al. Evaluation of the MicroScan colistin well and gradient diffusion strips for colistin susceptibility testing in Enterobacteriaceae. *J Clin Microbiol* 2019;57. <https://doi.org/10.1128/JCM.01866-18>.
- Martiny HM, Munk P, Brinch C et al. Global Distribution of mcr Gene Variants in 214K Metagenomic Samples. 2022, , 26, e0010522
- Matuschek E, Ahman J, Webster C et al. Antimicrobial susceptibility testing of colistin—evaluation of seven commercial MIC products against standard broth microdilution for *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and *Acinetobacter* spp. *Clin Microbiol Infect* 2018;24:865–70. <https://doi.org/10.1016/j.cmi.2017.11.020>.
- Mentasti M, David S, Sands K et al. Rapid detection and differentiation of mobile colistin resistance (mcr-1 to mcr-10) genes by real-time PCR and melt-curve analysis. *J Hosp Infect* 2021;110:148–55. <https://doi.org/10.1016/j.jhin.2021.01.010>.
- Mushtaq A, Nawaz H, Irfan Majeed M et al. Surface-enhanced Raman spectroscopy (SERS) for monitoring colistin-resistant and susceptible *E. coli* strains. *Spectrochim Acta A Mol Biomol Spectrosc* 2022;278:121315. <https://doi.org/10.1016/j.saa.2022.121315>.
- Nordmann P, Jayol A, Poirel L. A universal culture medium for screening polymyxin-resistant gram-negative isolates. *J Clin Microbiol* 2016b;54:1395–9. <https://doi.org/10.1128/JCM.00446-16>.
- Nordmann P, Jayol A, Poirel L. Rapid detection of polymyxin resistance in Enterobacteriaceae. *Emerg Infect Dis* 2016a;22:1038–43. <https://doi.org/10.3201/eid2206.151840>.
- Notomi T, Okayama H, Masubuchi H et al. Loop-mediated isothermal amplification of DNA. *Nucleic Acids Res* 2000;28:E63. <https://doi.org/10.1093/nar/28.12.e63>.
- Novovic K, Jovicic B. Colistin resistance in *Acinetobacter baumannii*: molecular mechanisms and epidemiology. *Antibiotics (Basel)* 2023;12:516.
- Osei Sekyere J, Sephofane AK, Mbelle NM. Comparative evaluation of CHROMagar COL-APSE, MicroScan Walkaway, ComASP Colistin, and Colistin MAC Test in detecting colistin-resistant Gram-negative bacteria. *Sci Rep* 2020;10:6221. <https://doi.org/10.1038/s41598-020-63267-2>.
- Park JW. Principles and applications of loop-mediated isothermal amplification to point-of-care tests. *Biosensors (Basel)* 2022;12:857.
- Przybylski SM, Correa-Martinez C, Kock R et al. SuperPolymyxin medium for the screening of colistin-resistant Gram-negative bacteria in stool samples. *Front Microbiol* 2018;9:2809. <https://doi.org/10.3389/fmicb.2018.02809>.
- Ragupathi NKD, Bakthavatchalam YD, Mathur P et al. Plasmid profiles among some ESKAPE pathogens in a tertiary care centre in south India. *Indian J Med Res* 2019;149:222–31.
- Ragupathi NKD, Sethuvel DPM, Anandan S et al. First hybrid complete genome of *Aeromonas veronii* reveals chromosome-mediated novel structural variant mcr-3.30 from a human clinical sample. *Access Microbiol* 2020;2:acmi000103. <https://doi.org/10.1099/acmi.0.000103>.
- Rebello AR, Bortolaia V, Kjeldgaard JS et al. Multiplex PCR for detection of plasmid-mediated colistin resistance determinants, mcr-1, mcr-2, mcr-3, mcr-4 and mcr-5 for surveillance purposes. *Euro Surveill* 2018;23. <https://doi.org/10.2807/1560-7917.ES.2018.23.6.17-00672>.
- Rodriguez-Manzano J, Moser N, Malpartida-Cardenas K et al. Rapid detection of mobilized colistin resistance using a nucleic acid based lab-on-a-chip diagnostic system. *Sci Rep* 2020;10:8448. <https://doi.org/10.1038/s41598-020-64612-1>.
- Saikia D, Jadhav P, Hole AR et al. Unraveling the secrets of colistin resistance with label-free Raman spectroscopy. *Biosensors (Basel)* 2022;12:749.
- Satlin MJ, Lewis JS, Weinstein MP et al. Clinical and Laboratory Standards Institute and European Committee on Antimicrobial Susceptibility Testing position statements on Polymyxin B and Colistin Clinical breakpoints. *Clin Infect Dis* 2020;71:e523–9.
- Schurch AC, van Schaik W. Challenges and opportunities for whole-genome sequencing-based surveillance of antibiotic resistance. *Ann NY Acad Sci* 2017;1388:108–20. <https://doi.org/10.1111/nyas.13310>.
- Sekyere JO. mcr colistin resistance gene: a systematic review of current diagnostics and detection methods. *Microbiol Open* 2019;8:e00682.
- Singh S, Pathak A, Rahman M et al. Genetic characterisation of colistin resistant *Klebsiella pneumoniae* clinical isolates from North India. *Front Cell Infect Microbiol* 2021;11:666030. <https://doi.org/10.3389/fcimb.2021.666030>.
- Smith RD, Izac JR, Ha M et al. Rapid identification of mcr-1-positive *Escherichia coli* from patient urine using a novel lipid-based MALDI-TOF-MS assay. *Access Microbiol* 2021;3:000309. <https://doi.org/10.1099/acmi.0.000309>.
- Smith RD, McElheny CL, Izac JR et al. A novel lipid-based MALDI-TOF assay for the rapid detection of colistin-resistant Enterobacter species. *Microbiol Spectr* 2022;10:e0144521. <https://doi.org/10.1128/spectrum.01445-21>.

- Tahmasebi H, Dehbashi S, Arabestani MR. Co-harboring of *mcr-1* and beta-lactamase genes in *Pseudomonas aeruginosa* by high-resolution melting curve analysis (HRMA): molecular typing of superbug strains in bloodstream infections (BSI). *Infect Genet Evol* 2020a;85:104518. <https://doi.org/10.1016/j.meegid.2020.104518>.
- Tahmasebi H, Dehbashi S, Arabestani MR. New approach to identify colistin-resistant *Pseudomonas aeruginosa* by high-resolution melting curve analysis assay. *Lett Appl Microbiol* 2020b;70:290–9. <https://doi.org/10.1111/lam.13270>.
- Tahmasebi H, Dehbashi S, Arabestani MR. Prevalence and molecular typing of colistin-resistant *Pseudomonas aeruginosa* (CRPA) among beta-lactamase-producing isolates: a study based on high-resolution melting curve analysis method. *Infect Drug Resist* 2020c;13:2943–55. <https://doi.org/10.2147/IDR.S264796>.
- Tolosi R, Apostolakis I, Laconi A *et al*. Rapid detection and quantification of plasmid-mediated colistin resistance genes (*mcr-1* to *mcr-5*) by real-time PCR in bacterial and environmental samples. *J Appl Microbiol* 2020;129:1523–9. <https://doi.org/10.1111/jam.14738>.
- Wang H, Chen Y, Strich JR *et al*. Rapid detection of colistin resistance protein MCR-1 by LC–MS/MS. *Clin Proteomics* 2019;16:8. <https://doi.org/10.1186/s12014-019-9228-2>.
- Wei C, Li M, Zhao X. Surface-enhanced Raman scattering (SERS) with silver nano substrates synthesized by microwave for rapid detection of foodborne pathogens. *Front Microbiol* 2018;9:2857. <https://doi.org/10.3389/fmicb.2018.02857>.
- Weiss R, Palatinszky M, Wagner M *et al*. Surface-enhanced Raman spectroscopy of microorganisms: limitations and applicability on the single-cell level. *Analyst* 2019;144:943–53. <https://doi.org/10.1039/C8AN02177E>.
- Zhang H, Yu F, Lu X *et al*. Rapid detection of MCR-mediated colistin resistance in *Escherichia coli*. *Microbiol Spectr* 2022;10:e0092022. <https://doi.org/10.1128/spectrum.00920-22>.
- Zhong LL, Zhou Q, Tan CY *et al*. Multiplex loop-mediated isothermal amplification (multi-LAMP) assay for rapid detection of *mcr-1* to *mcr-5* in colistin-resistant bacteria. *Infect Drug Resist* 2019;12:1877–87. <https://doi.org/10.2147/IDR.S210226>.